Exploratory Analysis of Rainfall Data in India for Agriculture

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Abstract

Rainfall is one of the most critical natural factors influencing agricultural productivity in India. Variations in rainfall patterns directly affect crop yield, irrigation planning, and agricultural sustainability. Understanding historical rainfall trends and patterns can help farmers, agricultural experts, and policymakers make informed decisions to reduce risks associated with droughts and floods.

This project focuses on the exploratory analysis of historical rainfall data across Indian subdivisions from 1901 to 2017. The dataset consists of monthly and annual rainfall measurements. Data preprocessing techniques were applied to handle missing values, convert data into numerical format, and prepare the dataset for analysis. Exploratory Data Analysis (EDA) was performed using visualization techniques such as trend analysis, monthly distribution charts, and correlation heatmaps to understand rainfall patterns.

A Random Forest Classifier was developed to classify heavy rainfall conditions based on monthly rainfall features. The model achieved an accuracy of 97%, demonstrating strong predictive performance. Furthermore, the trained model was deployed using the Flask web framework to create an interactive web application that allows users to input monthly rainfall values and obtain rainfall classification results.

This project demonstrates the practical application of Machine Learning and Web Deployment techniques in analyzing rainfall data for agricultural decision support systems.

1. Introduction

Agriculture is one of the most significant sectors of the Indian economy, employing a large portion of the population and contributing substantially to the country's GDP. The success of agriculture in India is highly dependent on rainfall patterns, especially during the monsoon season. Irregular or insufficient rainfall can lead to drought conditions, while excessive rainfall may cause floods, both of which severely impact crop production and farmer livelihoods.

Understanding rainfall patterns and trends over time is essential for effective agricultural planning, irrigation management, and risk mitigation. Traditional methods of rainfall analysis often rely on statistical summaries; however, with the advancement of Artificial Intelligence (AI) and Machine Learning (ML), it is possible to perform deeper analysis and develop predictive models based on historical weather data.

Exploratory Data Analysis (EDA) plays a crucial role in identifying rainfall trends, seasonal variations, and relationships among different monthly rainfall measurements. By analyzing historical rainfall data across Indian subdivisions, valuable insights can be extracted to support agricultural decision-making.

In this project, historical rainfall data from 1901 to 2017 is analyzed using data preprocessing and visualization techniques. A machine learning model is developed to classify heavy rainfall conditions based on monthly rainfall features. Additionally, the trained model is deployed using a Flask-based web application to provide an interactive prediction interface.

2. Problem Statement

Agriculture in India is highly dependent on rainfall, and variations in rainfall patterns significantly affect crop productivity, irrigation planning, and agricultural sustainability. Irregular rainfall can lead to droughts or floods, causing economic losses and food security challenges.

Although historical rainfall data is available, it is often not analyzed effectively to extract meaningful insights or develop predictive models. There is a need for a system that can analyze past rainfall patterns and classify heavy rainfall conditions based on monthly rainfall data.

The main objective of this project is to:

- Perform exploratory data analysis on historical rainfall data of Indian subdivisions.
- Identify rainfall trends and seasonal patterns.
- Develop a machine learning model to classify heavy rainfall conditions.
- Deploy the trained model using a web application for user interaction.

This system can assist in understanding rainfall behavior and support decision-making in agriculture-related planning.

3. Dataset Description

The dataset used in this project consists of historical rainfall records for various subdivisions of India. The data spans from the year 1901 to 2017 and includes detailed monthly as well as annual rainfall measurements.

The dataset contains a total of **4188 records**, where each record represents rainfall data for a particular subdivision and year.

The dataset includes the following attributes:

- **SUBDIVISION** – Name of the Indian subdivision
- **YEAR** – Year of rainfall measurement
- **JAN to DEC** – Monthly rainfall values (in millimeters)
- **ANNUAL** – Total annual rainfall

Each row in the dataset represents the rainfall distribution for a specific subdivision during a particular year.

The dataset provides sufficient historical information to analyze rainfall trends, seasonal variations, and relationships among different monthly rainfall values.

4. Data Preprocessing

Data preprocessing is a crucial step in any machine learning project. The rainfall dataset contained missing values and required cleaning before analysis and model development.
The following preprocessing steps were performed:

4.1 Handling Missing Values

Some rainfall values in the dataset were represented as "NA". These values were first converted into null (NaN) format.
After identifying missing values, **mean imputation** was applied to replace missing entries with the average value of the respective column.
This ensured that no data rows were removed and the dataset remained complete for model training.

```
df.fillna(df.mean(numeric_only=True), inplace=True)
```

```
                 SUBDIVISION  YEAR   JAN    FEB   MAR    APR
MAY   JUN  \
0  Andaman & Nicobar Islands  1901  49.2   87.1  29.2    2.3
528.8  517.5
1  Andaman & Nicobar Islands  1902   0.0  159.8  12.2    0.0
446.1  537.1
2  Andaman & Nicobar Islands  1903  12.7  144.0   0.0    1.0
235.1  479.9
3  Andaman & Nicobar Islands  1904   9.4   14.7   0.0  202.4
304.5  495.1
4  Andaman & Nicobar Islands  1905   1.3    0.0   3.3   26.9
279.5  628.7

     JUL    AUG    SEP    OCT    NOV    DEC  ANNUAL
0  365.1  481.1  332.6  388.5  558.2   33.6  3373.2
1  228.9  753.7  666.2  197.2  359.0  160.5  3520.7
2  728.4  326.7  339.0  181.2  284.4  225.0  2957.4
3  502.0  160.1  820.4  222.2  308.7   40.1  3079.6
4  368.7  330.5  297.0  260.7   25.4  344.7  2566.7
              YEAR            JAN            FEB            MAR
APR  \
count   4188.000000   4188.000000   4188.000000   4188.000000
4188.000000
mean    1959.220630     18.938337     21.599427     27.412912
43.071845
std       33.717097     33.776982     35.718931     46.894327
68.118424
min     1901.000000      0.000000      0.000000      0.000000
0.000000
```

25%	1930.000000	0.600000	0.500000	1.000000	3.000000
50%	1959.000000	6.000000	6.500000	7.900000	15.500000
75%	1988.000000	22.000000	26.600000	31.300000	49.525000
max	2017.000000	583.700000	403.500000	605.600000	595.100000

	MAY	JUN	JUL	AUG	SEP \
count	4188.000000	4188.000000	4188.000000	4188.000000	4188.000000
mean	85.660860	230.117475	347.020043	289.742185	197.316332
std	122.651174	234.152152	268.625870	188.336044	135.450491
min	0.000000	0.400000	0.000000	0.000000	0.100000
25%	8.675000	70.875000	175.900000	155.975000	100.400000
50%	36.900000	139.000000	285.250000	258.900000	173.900000
75%	97.725000	304.300000	418.400000	377.525000	265.825000
max	1168.600000	1609.900000	2362.800000	1664.600000	1222.000000

	OCT	NOV	DEC	ANNUAL
count	4188.000000	4188.000000	4188.000000	4188.000000
mean	95.324707	39.496433	18.968645	1409.449207
std	99.111897	68.268852	43.002315	899.792622
min	0.000000	0.000000	0.000000	62.300000
25%	14.600000	0.600000	0.100000	805.275000
50%	65.750000	9.450000	3.100000	1124.150000
75%	148.100000	45.000000	17.700000	1634.075000
max	948.300000	648.900000	617.500000	6331.100000

4.2 Data Type Conversion

All monthly rainfall columns (JAN–DEC) and the ANNUAL column were converted into numeric format to ensure proper mathematical operations and model compatibility.

4.3 Feature Selection

For building the machine learning model, only the monthly rainfall features were selected:

- JAN
- FEB
- MAR
- APR
- MAY
- JUN
- JUL
- AUG
- SEP
- OCT
- NOV
- DEC

The subdivision and year columns were not used as input features for prediction.

4.4 Target Variable Creation

A new classification variable called **Rain_Class** was created:

- Heavy Rainfall (1)
- Normal Rainfall (0)

This transformation allowed the problem to be treated as a **classification task** instead of regression.

5. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand rainfall trends, seasonal patterns, and relationships among monthly rainfall values. Visualization techniques were used to extract meaningful insights from the dataset.

5.1 Annual Rainfall Trend Analysis

A line graph was plotted to analyze the variation in annual rainfall over different years.

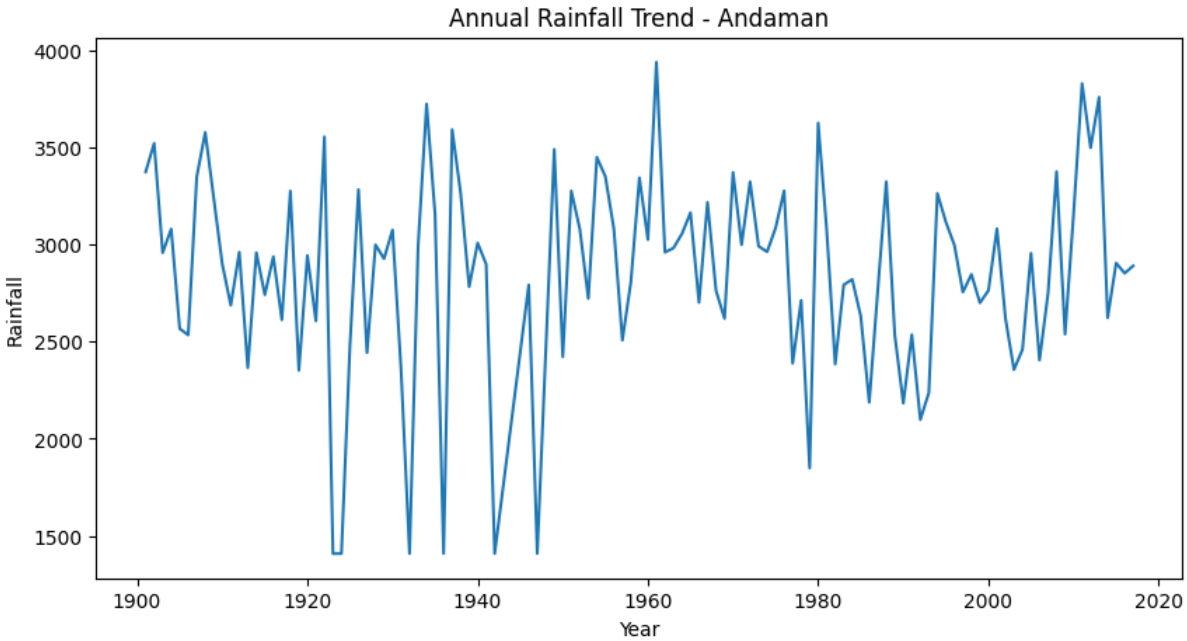

Figure 2: Annual Rainfall Trend Over Years

Now write this explanation:

The annual rainfall trend graph shows fluctuations in rainfall levels across different years. Some years experienced significantly higher rainfall, while others recorded lower rainfall values. This variation highlights the irregular nature of rainfall patterns in India, which can directly impact agricultural productivity.

5.2 Monthly Average Rainfall Distribution

A bar chart was created to visualize the average rainfall for each month.

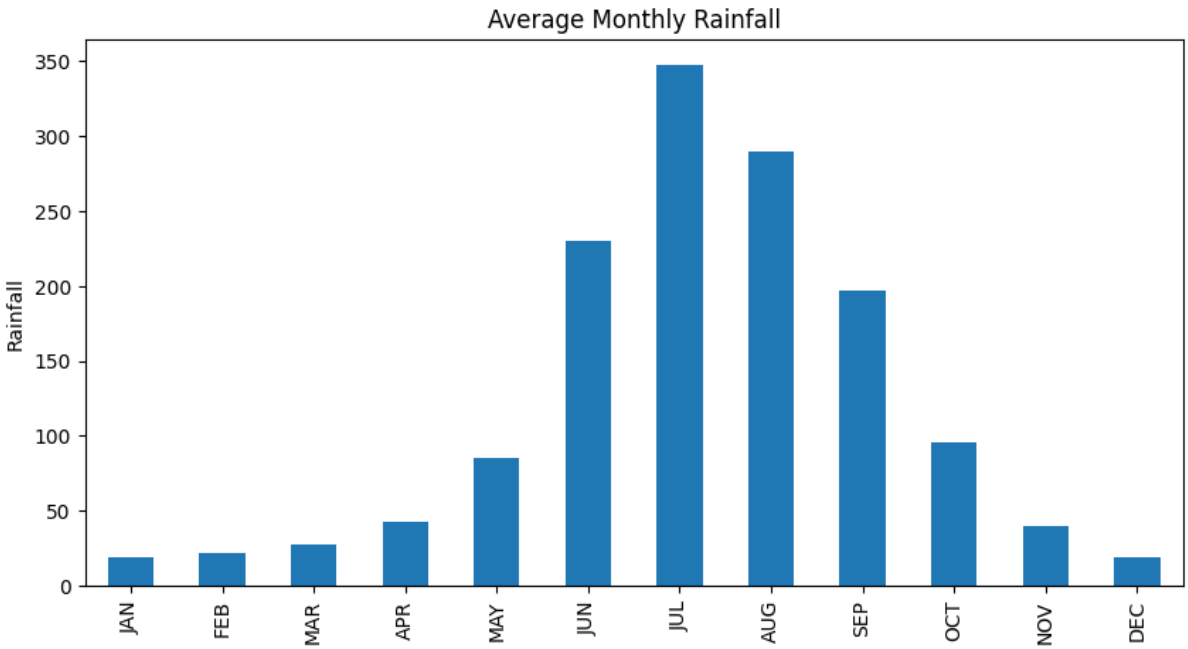

Figure 3: Average Monthly Rainfall Distribution

Now write this explanation:

The bar chart illustrates that rainfall is significantly higher during the monsoon months, particularly from June to September. These months contribute the highest proportion of annual rainfall. In contrast, months such as January, February, and December record comparatively lower rainfall levels.

5.3 Correlation Analysis

A correlation heatmap was generated to examine the relationships between monthly rainfall variables.

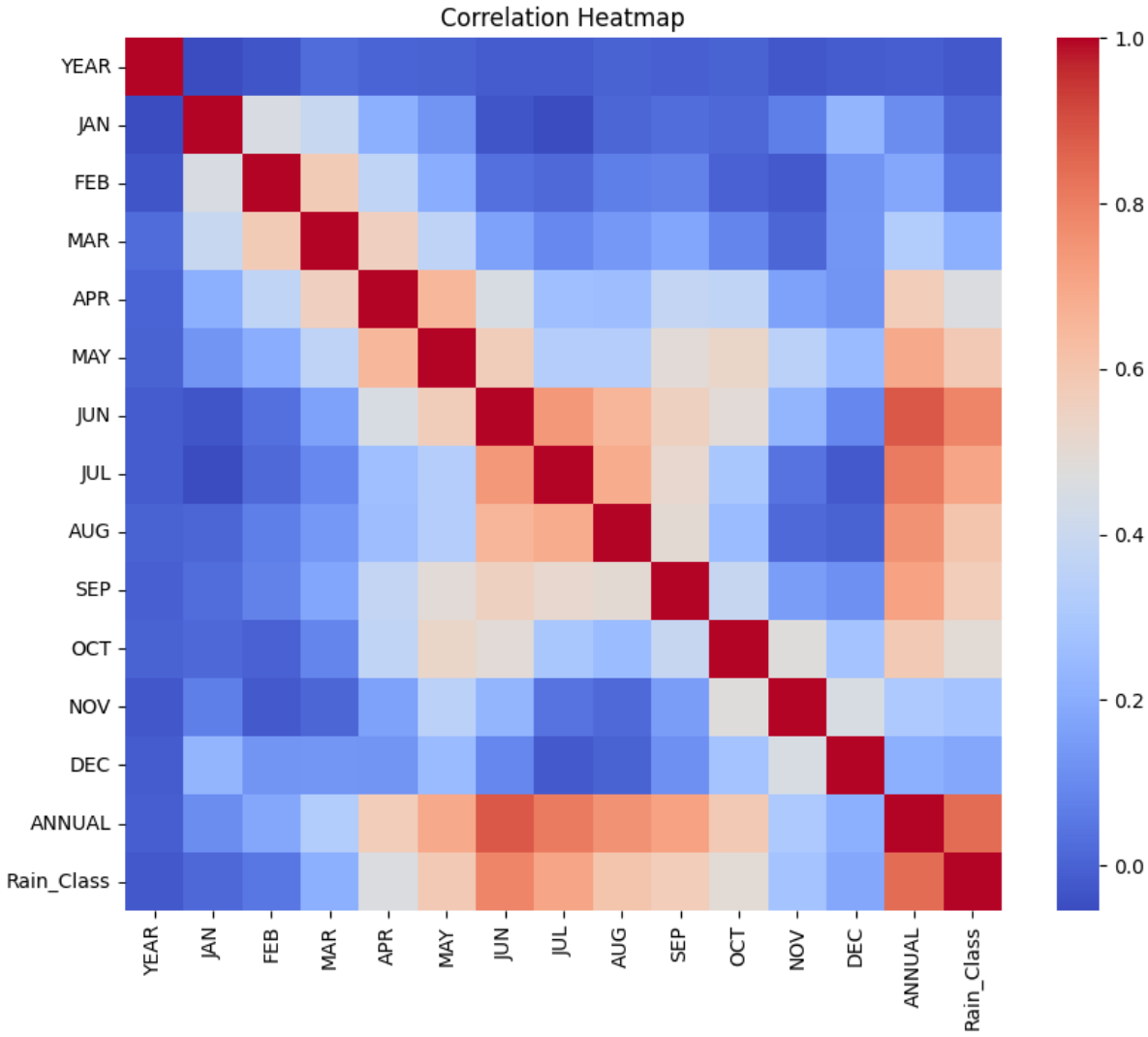

Figure 4: Correlation Matrix of Monthly Rainfall Features

Now write this explanation:

The correlation heatmap indicates strong positive correlations among monsoon months (June to September), suggesting that rainfall in these months tends to follow similar patterns. This confirms the seasonal behavior of rainfall distribution in India.

6. Model Building

After completing data preprocessing and exploratory analysis, a machine learning model was developed to classify rainfall conditions.

6.1 Problem Type Identification

The objective of the model was to classify whether the rainfall condition falls under:

- Heavy Rainfall (1)
- Normal Rainfall (0)

Since the output variable is categorical, this problem was treated as a **classification problem**.

6.2 Feature Selection

The following monthly rainfall columns were selected as input features:

- JAN
- FEB
- MAR
- APR
- MAY
- JUN
- JUL
- AUG
- SEP
- OCT
- NOV
- DEC

These features represent the rainfall distribution throughout the year.
The target variable used for classification was:

- **Rain_Class**

6.3 Train-Test Split

The dataset was divided into:

- 80% Training Data
- 20% Testing Data

This ensures that the model is trained on one portion of the data and tested on unseen data to evaluate performance.

6.4 Model Selection

The **Random Forest Classifier** algorithm was selected for this project because:

- It handles non-linear relationships effectively.
- It reduces overfitting through ensemble learning.
- It performs well on structured tabular data.

The model was trained using the training dataset and then tested on the testing dataset.

7. Model Evaluation

After training the Random Forest Classifier model, its performance was evaluated using the testing dataset.

7.1 Accuracy Score

The primary evaluation metric used in this project was **Accuracy**.
Accuracy measures the percentage of correct predictions made by the model.
The model achieved an accuracy of:
97%
This indicates that the model correctly classified rainfall conditions in 97 out of 100 cases, demonstrating strong predictive performance.

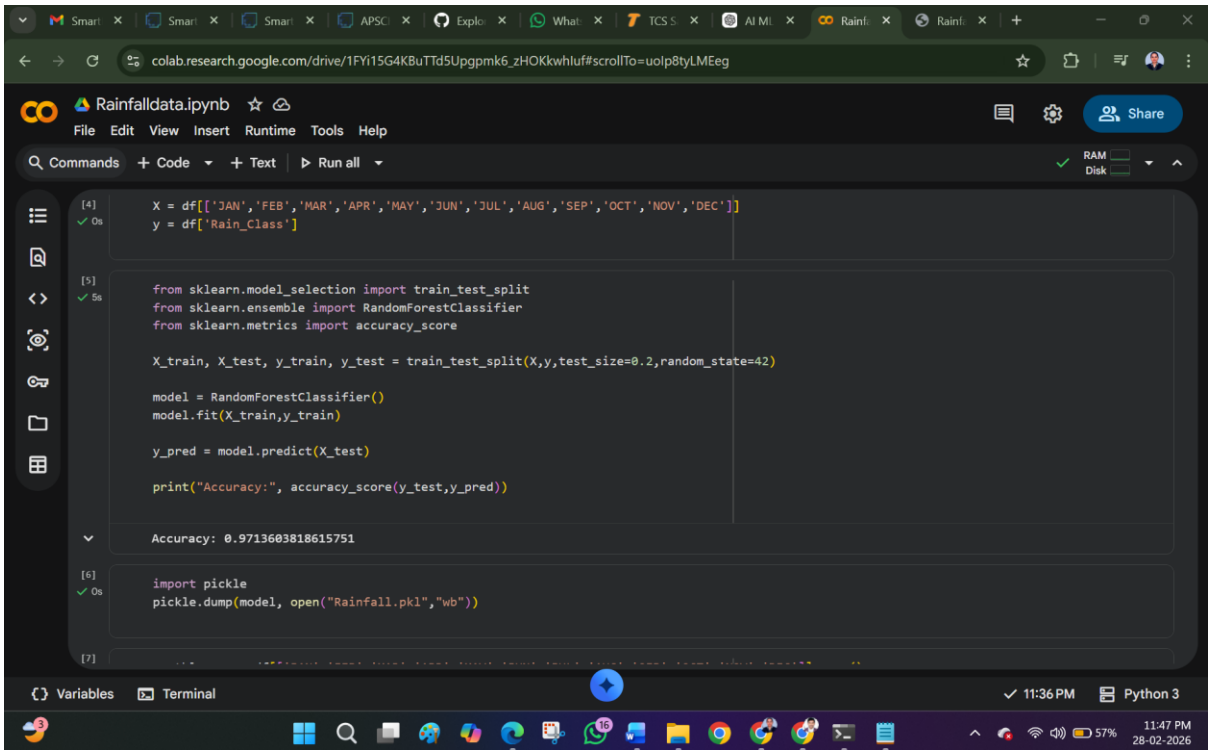

Figure 5: Model Accuracy Output

7.2 Interpretation of Results

The high accuracy value suggests that the selected features (monthly rainfall values) provide sufficient information to classify rainfall conditions effectively.
The Random Forest algorithm performed well due to:

- Its ability to handle complex feature relationships
- Its robustness against overfitting
- Ensemble-based decision making

The model shows strong reliability for rainfall classification based on historical data.

8. Web Application Deployment

After successfully training and evaluating the machine learning model, the next step was to deploy the model so that users can interact with it through a web interface.

8.1 Model Saving

The trained Random Forest model was saved using the **pickle** library.
Saving the model allows it to be reused without retraining every time the application runs.
Example:

```
import pickle
pickle.dump(model, open("Rainfall.pkl", "wb"))
The saved model file:
Rainfall.pkl
```

8.2 Flask Framework

The web application was developed using **Flask**, a lightweight Python web framework.
Flask was used to:

- Load the trained model
- Accept user input from a web form
- Process the input values
- Generate prediction output
- Display the result on the web page

8.3 Application Workflow

The web application follows this workflow:

1. User enters monthly rainfall values.
2. Input values are passed to the trained model.
3. The model predicts rainfall classification.
4. The result is displayed on the web interface.

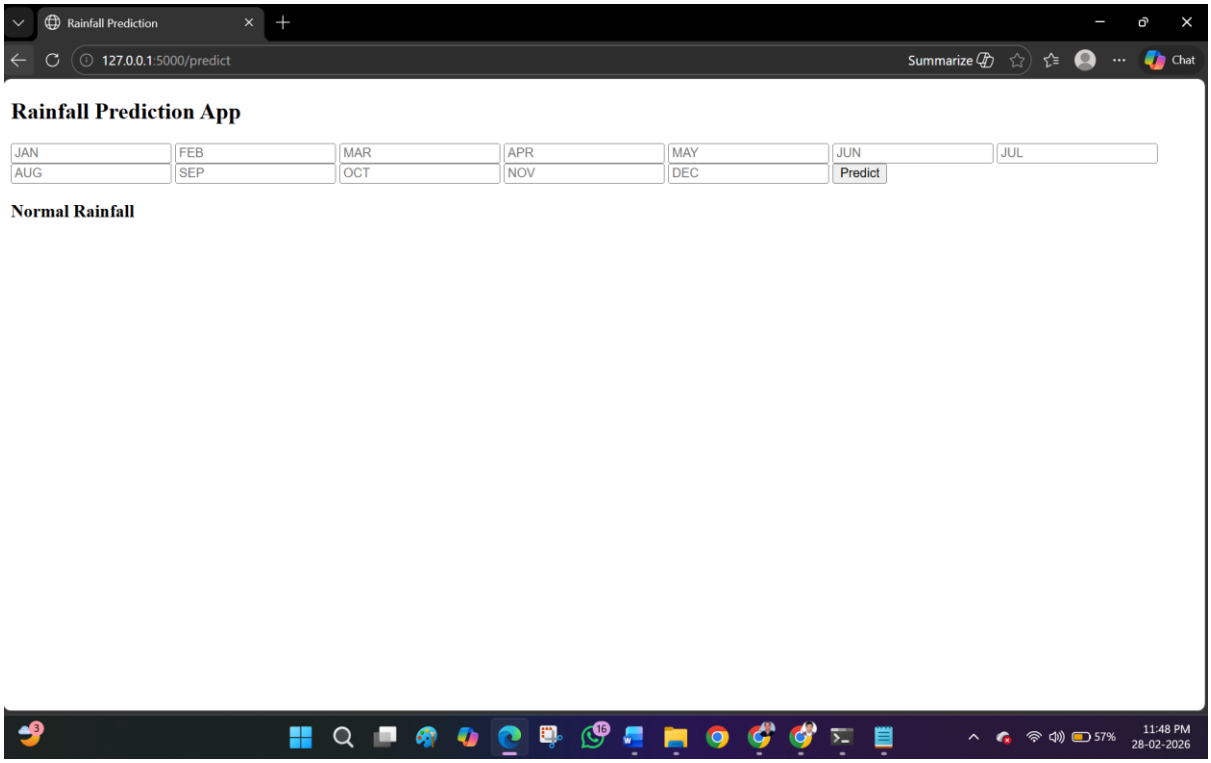

Figure 6: Rainfall Prediction Web Application Interface

8.4 Application Structure

The project structure includes:

- app.py – Flask application code
- Rainfall.pkl – Trained model
- templates/index.html – User interface
- Rainfall_prediction.ipynb – Model development notebook

This structure ensures proper separation of model, interface, and dataset components.

9. Conclusion

This project successfully performed exploratory analysis on historical rainfall data across Indian subdivisions from 1901 to 2017. The analysis helped in understanding rainfall trends, seasonal patterns, and correlations among monthly rainfall features.

Data preprocessing techniques were applied to handle missing values and prepare the dataset for modeling. A Random Forest Classifier was developed to classify heavy rainfall conditions based on monthly rainfall inputs. The model achieved an accuracy of **97%**, demonstrating strong predictive capability.

Furthermore, the trained model was deployed using the Flask web framework, allowing users to interact with the system through a web interface. This demonstrates the practical application of Machine Learning and Web Deployment techniques in agriculture-related decision support systems.

The project highlights how data-driven approaches can support better understanding of rainfall patterns and assist in agricultural planning.

10. Future Scope

Although the current system performs well, several improvements can be made in the future:

- Integration of real-time weather data using APIs.
- Implementation of advanced Deep Learning models for improved prediction.
- Deployment of the web application on cloud platforms such as AWS or Azure.
- Development of a mobile-based application for wider accessibility.
- Incorporation of additional environmental parameters such as temperature, humidity, and soil conditions.

These enhancements can further improve the system's reliability and usability for agricultural decision-making.